

Backend Engineering Fundamentals - 6 Week Intensive Course

Week 1: Python Fundamentals & Backend Basics

Learning Objectives

- Understand Python syntax and core programming concepts
- Learn about variables, data types, control flow, and functions
- Grasp the basics of what backend development entails
- Set up your development environment

Resources

Videos:

- [Python for Beginners - Full Course](#) by freeCodeCamp (4.5 hours)
- [What is Backend Development?](#) by Traversy Media (15 mins)
- [Python Functions Tutorial](#) by Corey Schafer (17 mins)

Text Resources:

- [Python Official Tutorial](#) - Chapters 3-6
- [Real Python - Python Basics](#)
- [Automate the Boring Stuff with Python](#) - Chapters 1-6

Topics Covered:

- Variables, data types (strings, integers, floats, booleans)
- Lists, tuples, dictionaries, sets
- Conditional statements (if/elif/else)
- Loops (for, while)
- Functions and parameters
- File handling basics

- Exception handling
- Python modules and imports

Week 1 Project: CLI Task Manager

Build a command-line task management application

Requirements:

- Add tasks with descriptions and due dates
- Mark tasks as complete
- View all tasks (pending and completed)
- Delete tasks
- Save tasks to a JSON file (persistence)
- Load tasks when the program starts

This project tests: functions, data structures (lists/dictionaries), file I/O, error handling, and user input validation.